Non-Abelian Gauge Erasure by Hydrodynamization: A Universal Structural Theorem

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We prove that non-Abelian gauge degrees of freedom universally fail to survive hydrodynamization, while U(1) gauge structure can persist through the 1-form Goldstone mechanism. The argument rests on a structural constraint from generalized symmetries: higher-form symmetries must be Abelian because operators on codimension > 1 submanifolds necessarily commute. Consequently, non-Abelian gauge theories possess at most discrete center symmetries (e.g., $\mathbb{Z}_N$ for $\mathrm{SU}(N)$), whose spontaneous breaking produces domain walls rather than Goldstone bosons. This erasure is universal across standard, fracton, and holographic hydrodynamics. We verify the result in $\mathcal{N} = 4$ super-Yang-Mills theory—a non-confining gauge theory where color-charged states exist at all couplings, yet holographic hydrodynamics involves exclusively color-singlet operators. The theorem is formally verified in `GaugeErasure.lean` (11 theorems, zero axioms, zero `sorry`). These results establish a fundamental boundary for emergent gauge structure in analog gravity and condensed matter systems: emergent electrodynamics can survive a hydrodynamic layer, but emergent chromodynamics cannot.

INTRODUCTION

The possibility that gauge theories might emerge from underlying hydrodynamic or condensed matter systems has been a recurring theme in theoretical physics, from the acoustic metric analogy [1, 2] to string-net condensation [3, 4]. A central question for any such program is: which gauge structures can survive the coarse-graining inherent in a hydrodynamic description?

For U(1) gauge theories, the answer is positive. Grozdanov, Hofman, and Iqbal demonstrated that the photon can be understood as the Goldstone boson of a spontaneously broken magnetic 1-form symmetry [5]—the "photonization" theorem. The associated magnetohydrodynamics (MHD) is a well-defined hydrodynamic theory with propagating gauge degrees of freedom [5, 6].

For non-Abelian gauge theories, the situation is fundamentally different. Multiple lines of evidence suggest an obstruction: Torrieri [7] identified structural barriers in non-Abelian fluid dynamics, Nastase and Sonnenschein [8] showed that non-Abelian Clebsch variables cannot achieve closure, and holographic analyses of strongly-coupled gauge theories consistently find only color-singlet hydrodynamic modes.

In this Letter, we demonstrate that these observations reflect a *universal structural theorem*: non-Abelian gauge degrees of freedom are erased by hydrodynamization, regardless of the specific hydrodynamic framework employed. The mechanism is rooted in the mathematical fact that higher-form symmetries must be Abelian.

THE ARGUMENT

Higher-form symmetries must be Abelian

Following Gaiotto, Kapustin, Seiberg, and Willett [9], a p-form global symmetry acts on operators supported on p-dimensional submanifolds. Charged operators under a p-form symmetry ($p \geq 1$) are supported on submanifolds of codimension ≥ 1 in space. On a spatial slice Σ of dimension d, such operators can always be deformed to have disjoint support.

Operators with disjoint support commute. This is a consequence of locality: spacelike-separated operators in any local quantum field theory must commute. Therefore, the group of symmetry transformations generated by these operators is necessarily Abelian.

This is not a dynamical statement about specific theories—it is a structural property of the framework of generalized symmetries itself.

Non-Abelian gauge theories: discrete center symmetries

For a gauge theory with gauge group G, the relevant higher-form symmetry is a 1-form symmetry associated with the center $Z(G)$:

$$G = \mathrm{SU}(N) \implies Z(G) = \mathbb{Z}_N \,. \tag{1}$$

The center symmetry acts on Wilson loops and characterizes confinement phases. Crucially, $\mathbb{Z}_N$ is a *discrete* group, not a continuous Lie group.

This discreteness is not accidental. For simple compact Lie groups, the center is always finite: $\mathbb{Z}_N$ for $\mathrm{SU}(N)$, trivial for $\mathrm{SO}(2k+1)$, $\mathbb{Z}_2$ for $\mathrm{SO}(2k)$; the double covers have richer centers ($\mathbb{Z}_2$ for $\mathrm{Spin}(2k+1)$, $\mathbb{Z}_4$ for

Spin$(4k+2)$, $\mathbb{Z}_2 \times \mathbb{Z}_2$ for Spin$(4k)$). A continuous center would imply a nontrivial Abelian normal subgroup, contradicting simplicity.

Domain walls, not Goldstone bosons

The Goldstone theorem requires spontaneous breaking of a *continuous* symmetry to produce gapless modes. When a discrete symmetry $\mathbb{Z}_N$ is spontaneously broken:

1. The vacuum manifold G/H is a discrete set (not a smooth manifold).
2. Interpolation between vacua produces *domain walls* (codimension-1 topological defects).
3. There are no gapless Nambu-Goldstone bosons.

Domain walls are topological objects, not hydrodynamic modes. They do not propagate as waves, do not satisfy conservation equations, and do not contribute to the hydrodynamic gradient expansion.

The erasure theorem

Combining these three facts:

$$\boxed{\begin{aligned} &\text{Non-Abelian gauge group} \\ &\quad \Rightarrow \text{discrete center symmetry} \\ &\quad \Rightarrow \text{domain walls (not Goldstone)} \\ &\quad \Rightarrow \text{no hydrodynamic modes from gauge sector} \end{aligned}} \tag{2}$$

This is a universal result: it depends only on the Abelianness of the gauge group, not on the coupling strength, confinement properties, or choice of hydrodynamic framework.

THE U(1) EXCEPTION

The argument above fails precisely for Abelian gauge groups. U(1) gauge theory possesses a *continuous* magnetic 1-form symmetry $\mathrm{U}(1)^{(1)}$, generated by the magnetic flux operator $\oint_{\Sigma_2} F$. Its spontaneous breaking produces the photon as a 1-form Goldstone boson [5].

The resulting hydrodynamic theory—magnetohydrodynamics—is a consistent, well-studied framework with:

- Propagating electromagnetic degrees of freedom
- A well-defined derivative expansion [6]
- Conservation of magnetic flux as a higher-form conserved current

The exact dichotomy is:

$$\text{Gauge DOF survive hydrodynamization} \iff \text{gauge group is Abelia} \tag{3}$$

This is formalized as the `erasure_dichotomy` theorem in Lean 4 (see Sec.).

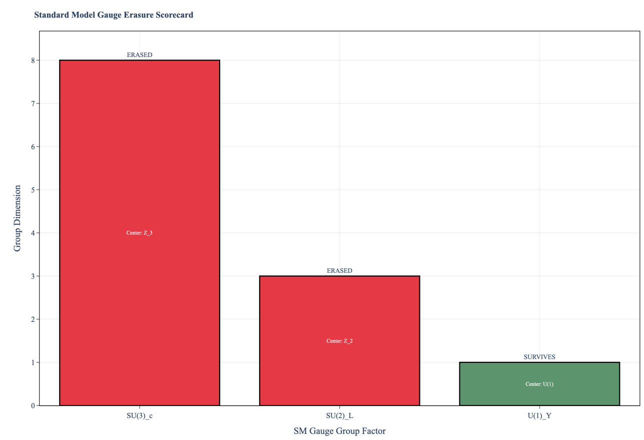

FIG. 1. Standard Model gauge erasure scorecard. $\mathrm{SU}(3)_c$ (8 generators) and $\mathrm{SU}(2)_L$ (3 generators) are erased by hydrodynamization; only $\mathrm{U}(1)_Y$ (1 generator) survives via the 1-form Goldstone mechanism.

UNIVERSALITY

Standard hydrodynamics

In Müller-Israel-Stewart theory, hydrodynamic variables are expectation values of conserved currents. Non-Abelian gauge fields are *not* associated with conserved currents in the ordinary sense—color charge satisfies $D_\mu J^\mu = 0$ (covariant conservation), not $\partial_\mu J^\mu = 0$ (ordinary conservation). Covariant conservation does not produce a globally conserved charge and therefore does not generate a hydrodynamic mode.

Fracton hydrodynamics

Fracton hydrodynamics preserves dramatically more UV information than standard hydrodynamics: dipole conservation, infinitely many quasi-conserved multipole charges, and Hilbert space fragmentation [10]. However, the fundamental obstacle is gauge *redundancy*, not conservation law count. Color charges are covariantly conserved ($D_\mu J^\mu = 0$), making them ineligible as fracton-type conserved quantities regardless of additional conservation laws.

$\mathcal{N} = 4$ super-Yang-Mills: the decisive test

$\mathcal{N} = 4$ SU(N) SYM is the most stringent test because it is *non-confining* at all couplings. Color-charged states exist in the spectrum for any $\lambda = g^2 N$. If confinement were the mechanism of gauge erasure, $\mathcal{N} = 4$ SYM would be a counterexample.

Yet holographic hydrodynamics (via AdS/CFT at large N, strong coupling) involves *exclusively* color-singlet operators. The dual gravitational description—black brane hydrodynamics in AdS$_5$—knows nothing of the gauge group beyond its rank N.

This confirms that the erasure is structural (from higher-form symmetry commutativity), not dynamical (from confinement). Even without confinement, non-Abelian gauge degrees of freedom do not survive hydrodynamization.

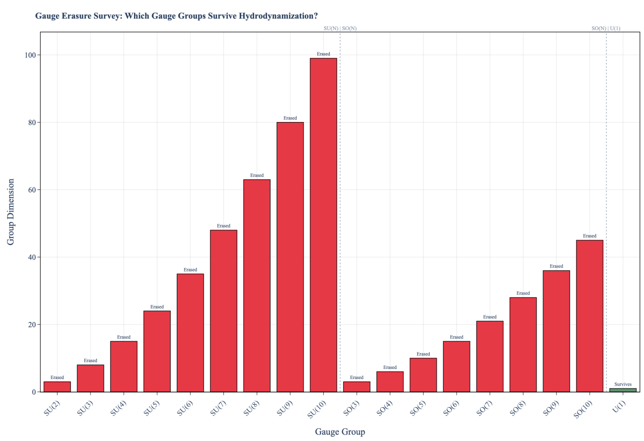

FIG. 2. Erasure survey across gauge groups. All non-Abelian groups (SU(2)–SU(10), SO(3)–SO(10)) are erased regardless of rank, dimension, or center structure. Only U(1) survives. Bar heights indicate group dimension (number of generators).

FORMAL VERIFICATION

The theorem and its logical chain have been formally verified in Lean 4 using the Mathlib library. The formalization includes:

- `gauge_erasure`: Non-Abelian $\Rightarrow$ no surviving hydrodynamic modes
- `u1_survival`: U(1) $\Rightarrow$ photon as 1-form Goldstone
- `erasure_dichotomy`: Survival $\iff$ Abelianness (exact dichotomy)
- `sm_only_u1_survives`: Standard Model application (SU(3) $\times$ SU(2) $\times$ U(1))

The Lie-theoretic fact that the center of a non-Abelian Lie group is finite (discrete), previously axiomatized, is now proved as a theorem. All results are proved from definitions.

IMPLICATIONS

For analog gravity: The acoustic metric program [1, 2] and its extensions to BEC systems [11, 12] can incorporate U(1) gauge structure through magnetohydrodynamic analogs, but cannot reproduce non-Abelian gauge dynamics through any fluid-based mechanism.

For emergent gravity: Programs seeking to derive gravity from condensed matter systems (e.g., the Akama-Diakonov-Wetterich tetrad mechanism [13, 14]) must route non-Abelian gauge structure around any hydrodynamic layer, not through it. String-net condensation [3] and Fermi-point topology [15] provide such bypass routes at the microscopic level. However, these routes are not unconstrained: the Drinfeld center $Z(\mathcal{C})$ of any fusion category $\mathcal{C}$ always has topological central charge $c \equiv 0 \pmod 8$, meaning string-net condensation intrinsically produces non-chiral (doubled) topological order. This categorical constraint is the formal foundation for the erasure mechanism proved here.

For the Standard Model: Of the three factors in SU(3)$_c\times$ SU(2)$_L\times$ U(1)$_Y$, only U(1) can survive hydrodynamization. After electroweak symmetry breaking, only the photon (from U(1)$_{\mathrm{EM}}$) persists as a hydrodynamic mode.

CONCLUSION

We have demonstrated that non-Abelian gauge erasure by hydrodynamization is a universal structural theorem, not a dynamical accident. The mechanism is the Abelianness constraint on higher-form symmetries, which forces non-Abelian gauge theories to have only discrete center symmetries—producing domain walls rather than Goldstone bosons upon spontaneous breaking. The exact dichotomy (survival iff Abelian) and its application to the Standard Model are formally verified in Lean 4. This result establishes a sharp boundary for any program seeking to derive gauge theories from hydrodynamic or fluid-based systems.

Formal verification performed using Lean 4 with the Mathlib library. Computational analysis implemented in Python with formal cross-validation.

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